WORKED SOLUTION

STEP II 2025, Question 9

Mechanics • Collisions and Projectiles

STEP II 2025, Question 9

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2025 STEP papers (a zip file):

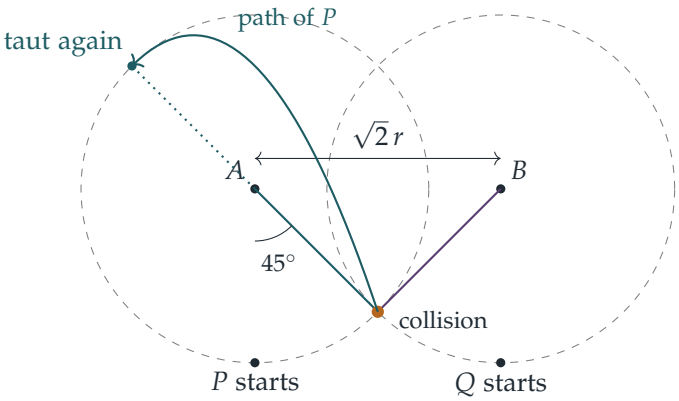
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The key idea

Symmetry does most of the work. The particles must meet on the perpendicular bisector of AB, which fixes where they are and how fast they move when they collide. Just after the collision each string goes slack, so each particle is an ordinary projectile until its distance from the point of suspension is back up to r. Writing that distance condition out gives a quartic in t. A triple root has a physical meaning: the particle reaches the circle travelling along it, so the string tightens without a jerk.

Solution

Model: P and Q are particles of mass m (small, smooth spheres for the collision), the strings are light and inextensible, and there is no air resistance. Put P on the left, hanging from A, and take axes through A with x horizontal (towards B) and y vertically up.



The figure shows P's flight after the collision in the special case of part (iv), where it rejoins its circle at the point 45° above the horizontal on the far side.

Part (i)

The set-up is symmetric about the perpendicular bisector of AB, so the particles meet on it. Each has then moved horizontally $\frac{1}{2}\sqrt{2}\,r = r\sin 45^\circ$ from its lowest point, so each string makes 45° with the downward vertical. Each particle has risen by $r - r\cos 45^\circ = r\left(1 - \frac{1}{\sqrt{2}}\right)$. The string is taut up to this point and its tension does no work, so by conservation of energy the speed v just before impact satisfies

$$v^2 = u^2 - 2gr\left(1 - \tfrac{1}{\sqrt{2}}\right) = u^2 - gr(2 - \sqrt{2}).$$

(We need $u^2 > gr(2-\sqrt{2})$ for the particles to get this far.)

Each particle is moving along its circle, at 45° to the horizontal, so

$$\text{velocity of } P = \left(\tfrac{v}{\sqrt{2}},\ \tfrac{v}{\sqrt{2}}\right), \qquad \text{velocity of } Q = \left(-\tfrac{v}{\sqrt{2}},\ \tfrac{v}{\sqrt{2}}\right).$$

The particles meet at the same height, so the line of centres is horizontal and the impulse between them is horizontal. Vertical components are therefore unchanged. Horizontally, the momenta are equal and opposite, so the total is zero both before and after. Newton's law of restitution then makes each particle rebound with e times its approach speed. Immediately after the collision,

$$\text{horizontal speed} = \frac{ev}{\sqrt{2}} = \tfrac{1}{2}ev\sqrt{2}, \qquad \text{vertical component} = \tfrac{1}{2}\sqrt{2}\,v \text{ (upwards)},$$

with P now moving away from Q, to the left.

Ask yourself Could the strings exert an impulse during the collision as well?

No. A string can only pull, and only along itself. Just after the collision, P is at $\left(\frac{r}{\sqrt{2}}, -\frac{r}{\sqrt{2}}\right)$ relative to A with velocity $\left(-\frac{ev}{\sqrt{2}}, \frac{v}{\sqrt{2}}\right)$. Its component along $\overrightarrow{AP}$ is $-\frac{1}{2}(1+e)v < 0$, so P is moving towards A. The string goes slack straight away, and an impulsive tension is never needed.

Part (ii)

Measure t from the collision. Until its string is taut again, P is a projectile, so relative to A

$$x = \frac{r}{\sqrt{2}} - \frac{ev}{\sqrt{2}}\,t, \qquad y = -\frac{r}{\sqrt{2}} + \frac{v}{\sqrt{2}}\,t - \tfrac{1}{2}gt^2.$$

The string is taut exactly when $x^2 + y^2 = r^2$. Multiplying by 2, so that $\sqrt{2}\,x$ and $\sqrt{2}\,y$ appear, gives

$$(r - evt)^2 + \left(-r + vt - \tfrac{1}{2}\sqrt{2}\,gt^2\right)^2 = 2r^2.$$

This holds at $t = 0$, when the string is still taut at the instant of collision, so the time we want is a non-zero root. By symmetry Q does the same at the same moment. The two particles move apart horizontally, so they cannot collide again first.

Part (iii)

Expanding and cancelling the $2r^2$:

$$\tfrac{1}{2}g^2t^4 - \sqrt{2}\,gv\,t^3 + \big((1+e^2)v^2 + \sqrt{2}\,rg\big)t^2 - 2(1+e)rv\,t = 0.$$

Dividing by $t \neq 0$ leaves a cubic. Now

$$\frac{z}{c} = \frac{vt}{r}\cdot\frac{rg}{\sqrt{2}\,v^2} = \frac{gt}{\sqrt{2}\,v}, \qquad \text{so} \qquad t = \frac{\sqrt{2}\,v}{g}\cdot\frac{z}{c}.$$

Substitute this, and divide the whole cubic by $\sqrt{2}\,v^3/g$. Writing $w = z/c$, the terms become

$$\begin{aligned}
\tfrac{1}{2}g^2t^3 &\longrightarrow w^3,\\
-\sqrt{2}\,gv\,t^2 &\longrightarrow -2w^2,\\
\big((1+e^2)v^2 + \sqrt{2}\,rg\big)t &\longrightarrow \Big(1+e^2+\frac{\sqrt{2}\,rg}{v^2}\Big)w = \Big(\frac{2}{c}+1+e^2\Big)w,\\
-2(1+e)rv &\longrightarrow -\frac{\sqrt{2}\,rg}{v^2}(1+e) = -\frac{2}{c}(1+e),
\end{aligned}$$

using $\dfrac{2}{c} = \dfrac{2rg}{\sqrt{2}\,v^2} = \dfrac{\sqrt{2}\,rg}{v^2}$. This is the required equation

$$\Big(\frac{z}{c}\Big)^3 - 2\Big(\frac{z}{c}\Big)^2 + \Big(\frac{2}{c}+1+e^2\Big)\Big(\frac{z}{c}\Big) - \frac{2}{c}(1+e) = 0.$$

Part (iv)

If the cubic in w has a triple root α, it must be $(w-\alpha)^3 = w^3 - 3\alpha w^2 + 3\alpha^2 w - \alpha^3$. Comparing coefficients,

$$3\alpha = 2, \qquad 3\alpha^2 = \frac{2}{c}+1+e^2, \qquad \alpha^3 = \frac{2}{c}(1+e).$$

So $\alpha = \frac{2}{3}$. Write $k = 2/c$. Then $k = \frac{4}{3} - 1 - e^2 = \frac{1}{3} - e^2$, and the last equation becomes

$$\Big(\tfrac{1}{3} - e^2\Big)(1+e) = \tfrac{8}{27} \iff 27e^3 + 27e^2 - 9e - 1 = 0 \iff (3e-1)(9e^2+12e+1) = 0.$$

The quadratic factor has roots $\frac{-2\pm\sqrt{3}}{3}$, both negative, so they cannot be coefficients of restitution. Hence $e = \frac{1}{3}$. Then $k = \frac{1}{3} - \frac{1}{9} = \frac{2}{9}$, so $c = 9$, and

$$v^2 = \frac{c\,rg}{\sqrt{2}} = \frac{9rg}{\sqrt{2}} = \tfrac{9}{2}\sqrt{2}\,rg,$$

as required.

? **Ask yourself** What does a repeated root say about the motion?

Let $D(t) = x^2 + y^2 - r^2$. Up to a positive factor, D equals $w(w-\alpha)^3$. This is negative for $0 < w < \alpha$, so P stays strictly inside its circle until the string tightens. Because α is a repeated root, $D' = 0$ there too. But $D' = 2(x\dot{x} + y\dot{y})$, so the component of velocity along the string is zero at that moment: the particle arrives moving along the circle. An impulsive tension would act along the string and would only be needed to remove a component of velocity along it. Here there is no such component, so there is no impulse, the velocity does not change, and no kinetic energy is lost.

Part (v)

Energy is lost only in the collision. Between the collision and the bottom of the circle, only gravity does work: during flight there is no tension, the tightening is jerk-free, and afterwards the tension is perpendicular to the motion. Just after the collision, the speed squared is

$$\frac{e^2v^2}{2} + \frac{v^2}{2} = \frac{(1+e^2)v^2}{2},$$

and the particle is $r\left(1 - \frac{1}{\sqrt{2}}\right)$ above its lowest point. Therefore

$$\begin{aligned} U^2 &= \frac{(1+e^2)v^2}{2} + gr\left(2-\sqrt{2}\right) = \frac{1}{2}\cdot\frac{10}{9}\cdot\frac{9rg}{\sqrt{2}} + gr\left(2-\sqrt{2}\right) \\ &= \frac{5\sqrt{2}}{2}rg + (2-\sqrt{2})rg = \left(2+\tfrac{3}{2}\sqrt{2}\right)rg = \frac{4+3\sqrt{2}}{2}rg. \end{aligned}$$

We should also check that the string stays taut on the way down. With $w = \frac{2}{3}$, $t = \frac{2\sqrt{2}v}{3g}$. Substituting gives position $\left(-\frac{r}{\sqrt{2}}, \frac{r}{\sqrt{2}}\right)$ and velocity $\left(-\frac{v}{3\sqrt{2}}, -\frac{v}{3\sqrt{2}}\right)$. So P rejoins its circle 45° above the horizontal, on the side away from B, with speed squared $\frac{v^2}{9} = \frac{rg}{\sqrt{2}}$. Resolving towards A there,

$$T + mg\sin 45^\circ = \frac{m}{r}\cdot\frac{rg}{\sqrt{2}} \quad\Longrightarrow\quad T = 0,$$

so the string becomes taut with zero tension. As P swings down, its speed increases and the inward component of gravity decreases, so $T > 0$ from then on. The particle follows the circle down to its lowest point, and the same value of U^2 comes from energy measured from this point: $\frac{rg}{\sqrt{2}} + 2gr\left(1+\frac{1}{\sqrt{2}}\right) = \left(2+\frac{3}{2}\sqrt{2}\right)rg$.

Checks

- With $r = 1$ and $g = 9.81$, a step-by-step computer simulation of the flight and the circular swing gave $U^2 \approx 40.430$. This agrees with $\frac{1}{2}(4+3\sqrt{2})g \approx 40.430$, and the tension stayed non-negative throughout the swing.
- The two routes to U^2 (from the collision point, and from the point where the string becomes taut) agree.
- If $e = 1/3$ and $c = 9$, then $w^3 - 2w^2 + \frac{4}{3}w - \frac{8}{27} = \left(w - \frac{2}{3}\right)^3$ exactly.

Common slips

- Taking the height risen as $r\cos 45^\circ$ (the depth below the point of suspension) rather than $r - r\cos 45^\circ$.
- Applying restitution to the whole velocity. Only the component along the line of centres (here horizontal) is affected.
- Forgetting to rule out the negative roots of the cubic in e.

Beyond STEP

The repeated-root condition says that two curves (the projectile's parabola and the circle) touch to high order: they meet, share a tangent, and here even share curvature, which is why the tension is exactly zero when the string becomes taut. Order of contact and osculating circles appear in first-year courses on curves and differential geometry. Impulsive constraints, and when they dissipate energy, are a standard topic in dynamics. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good place to see these ideas developed.

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